Homework 8: Object Oriented Programming Part 1

Texas Hold'em Poker: card game for 2–6 players in which players attempt to make the best possible 5-card hand using a combination of 2 private given cards and 5 shared community cards placed face up on the table.

Objects

Playing Card, Deck, Hand, Player, Pot, Hand Evaluator, Poker Game

Classes

Card, Deck, Hand, Player, Pot, HandEvaluator, PokerGame, Runner

Uses

- Deck has Cards
- Player has Hand
- Hand has list of Cards
- PokerGame has Deck
- PokerGame has Pot
- PokerGame has list of Players
- PokerGame has the list of community Cards (board)
- PokerGame has a Scanner (passed in from Runner)
- Runner creates Players and Scanner, then constructs and starts PokerGame

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Card

Attributes

Name	Visibility	Type	Description
rank	private	int	Numeric rank of the card (2–14, where 11=J, 12=Q, 13=K, 14=A)
suit	private	String	Suit of the card: "♥", "♠", "♣", or "♦"
rankLetter	private	String	String representation of the rank, set during toString()
faceCards	private	String[]	Lookup array {"J", "Q", "K", "A"} for ranks 11–14

Methods

Method	Parameters	Return	Description
Card(int rank, String suit)	rank (int), suit (String)	void (constructor)	Initializes a card with the given rank and suit
getRank()	none	int	Returns the card's numeric rank
getSuit()	none	String	Returns the card's suit symbol
toString()	none	String	Returns a readable string like "A-♠" or "10-♥" for display
toLines()	none	String[]	Returns a 7-line ASCII-art representation of the card
renderCards(ArrayList<Card> cards)	cards (ArrayList<Card>)	String (static)	Renders multiple cards side-by-side as ASCII art

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Deck

Attributes

Name	Visibility	Type	Description
cards	private	ArrayList<Card>	The list of all cards currently remaining in the deck
SUITS	private final	String[]	The four suit symbols: {"♥", "♠", "♣", "♦"}

Methods

Method	Parameters	Return	Description
Deck()	none	void (constructor)	Creates and populates a full 52-card deck
shuffle()	none	void	Randomizes card order using Math.random()-based Fisher–Yates shuffle
dealCard()	none	Card	Removes and returns the last card from the deck

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Hand

Attributes

Name	Visibility	Type	Description
holeCards	private	ArrayList<Card>	The two private cards dealt to the player
handValue	private	int	Numeric rank of the best 5-card combination (0=High Card … 9=Royal Flush)
yourBestHand	private	ArrayList<Card>	The best 5-card combination selected from hole cards + community cards

Methods

Method	Parameters	Return	Description
Hand()	none	void (constructor)	Initializes an empty hand with handValue = 0
addCard(Card c)	c (Card)	void	Adds a card to the hole cards list
clearHand()	none	void	Resets holeCards, handValue, and yourBestHand at the start of each new round
setHandValue(int value, ArrayList<Card> best)	value (int), best (ArrayList<Card>)	void	Stores the evaluated hand value and best 5-card combination
getHandValue()	none	int	Returns the hand's numeric value
getHoleCards()	none	ArrayList<Card>	Returns the player's private hole cards
getYourBestHand()	none	ArrayList<Card>	Returns the best 5-card combination for showdown display

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Player

Attributes

Name	Visibility	Type	Description
name	private	String	Display name of the player
chips	private	int	Current chip stack

Name	Visibility	Type	Description
hand	private	Hand	The player's current Hand object
currentBet	private	int	Amount wagered in the current betting round
folded	private	boolean	True if the player has folded this hand
allIn	private	boolean	True if the player has committed all remaining chips

Methods

Method	Parameters	Return	Description
Player(String name, int chips)	name (String), chips (int)	void (constructor)	Sets the player's name and starting chip count
getName()	none	String	Returns the player's name
getChips()	none	int	Returns current chip count
getPlayerHand()	none	Hand	Returns the player's Hand object
getCurrentBet()	none	int	Returns the player's current bet for this street
isFolded()	none	boolean	Returns whether the player has folded
isAllIn()	none	boolean	Returns whether the player is all-in
placeBet(int amount)	amount (int)	int	Deducts chips, updates currentBet, sets allIn if chips reach 0, returns amount
fold()	none	void	Sets folded to true
receiveChips(int amount)	amount (int)	void	Adds chips to the player's stack when they win a pot
resetNewRound()	none	void	Resets currentBet, folded, allIn, and clears the hand
resetBet()	none	void	Resets currentBet to 0 (used between streets within a hand)
makeDecision(int potCurrentBet, Pot pot, Scanner scanner)	potCurrentBet (int), pot (Pot), scanner (Scanner)	void	Prompts the player via Scanner for an action (fold, call, check, raise), validates input, and executes the action directly

Pot

Attributes

Name	Visibility	Type	Description
totalChips	private	int	Total chips currently in the pot
currentBet	private	int	The highest bet in the current round that others must match

Methods

Method	Parameters	Return	Description
Pot()	none	void (constructor)	Initializes totalChips and currentBet to 0
addChips(int amount)	amount (int)	void	Adds chips to the pot total
setCurrentBet(int bet)	bet (int)	void	Updates the current round's highest bet
getCurrentBet()	none	int	Returns the current round bet amount
getPotSize()	none	int	Returns total chips in the pot
reset()	none	void	Resets the pot to 0 at the start of each new hand

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HandEvaluator

Attributes

Name	Visibility	Type	Description
HAND_NAMES	public static final	String[]	Maps hand rank integers (0–9) to names: "High Card", "One Pair", "Two Pair", "Three of a Kind", "Straight", "Flush", "Full House", "Four of a Kind", "Straight Flush", "Royal Flush"

Methods

Method	Parameters	Return	Description
evaluate(Hand hand, ArrayList<Card> community)	hand (Hand), community (ArrayList<Card>)	int	Combines hole cards + community, tests all 21 five-card subsets, sets the best hand value on the Hand object, returns the rank integer

Method	Parameters	Return	Description
handValue(ArrayList<Card> five)	five (ArrayList<Card>)	int (private)	Scores a single 5-card hand (0–9) using frequency counts, flush detection, and straight detection
isStraight(int[] freq)	freq (int[])	boolean (private)	Returns true if the frequency array contains 5 consecutive ranks (handles A-low wheel)
getSortedRanks(ArrayList<Card> cards)	cards (ArrayList<Card>)	int[] (private)	Returns ranks sorted high-to-low via insertion sort
getCompareKey(ArrayList<Card> hand, int handRank)	hand (ArrayList<Card>), handRank (int)	int[] (private)	Builds a comparison key array ordered by importance (e.g., trips rank before kickers) for tiebreaking
compareHands(ArrayList<Card> a, ArrayList<Card> b, int handRank)	a (ArrayList<Card>), b (ArrayList<Card>), handRank (int)	int	Compares two best-five-card hands of the same rank; returns positive if a wins, negative if b wins, 0 for tie

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PokerGame

Attributes

Name	Visibility	Type	Description
players	private	ArrayList<Player>	All players currently in the game
deck	private	Deck	The deck used for the current hand
pot	private	Pot	The pot for the current hand
community	private	ArrayList<Card>	The shared board cards (Flop, Turn, River)
dealer	private	int	Index of the current dealer in the players list
smallBlind	private	int	Chip value of the small blind
bigBlind	private	int	Chip value of the big blind (always 2× smallBlind)
scanner	private	Scanner	Shared Scanner instance for player input

Methods

Method	Parameters	Return	Description
PokerGame(ArrayList<Player> players, int smallBlind, Scanner scanner)	players (ArrayList<Player>), smallBlind (int), scanner (Scanner)	void (constructor)	Initializes game state; receives pre-built player list and Scanner from Runner
startGame()	none	void	Outer loop: plays hands, eliminates broke players, rotates dealer until one player remains
playHand(int smallBlindIdx)	smallBlindIdx (int)	void	Runs a single hand: new deck, shuffle, blinds, deal, four betting rounds (pre-flop/flop/turn/river), showdown
dealHoleCards()	none	void	Deals 2 cards to each player
dealCommunityCards(int num)	num (int)	void	Burns one card, then deals num cards to the community board (3 for Flop, 1 for Turn/River)
bettingRound(int startIdx)	startIdx (int)	void	Iterates active players from startIdx, calls makeDecision(), updates pot; loops until no re-raises
showdown()	none	void	Evaluates all non-folded hands, breaks ties via compareHands, awards pot to winner(s), handles splits
postBlinds(int smallBlindIdx)	smallBlindIdx (int)	void	Forces small and big blind bets, adds to pot, sets pot's current bet
eliminateBroke()	none	void	Removes players with 0 or fewer chips from the game
displayCommunity()	none	void	Prints community cards as ASCII art (skips if empty)
resetStreet()	none	void (private)	Resets pot's current bet to 0 and each player's currentBet to 0 between streets

Runner

Methods

Method	Parameters	Return	Description
main(String[] args)	args (String[])	void (static)	Entry point: creates a Scanner, prompts for number of players (2–6), starting chips, small blind amount, and player names; constructs the ArrayList<Player> and PokerGame, then calls startGame()